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Course Code: ITA0330
Course Name: Mobile Computing

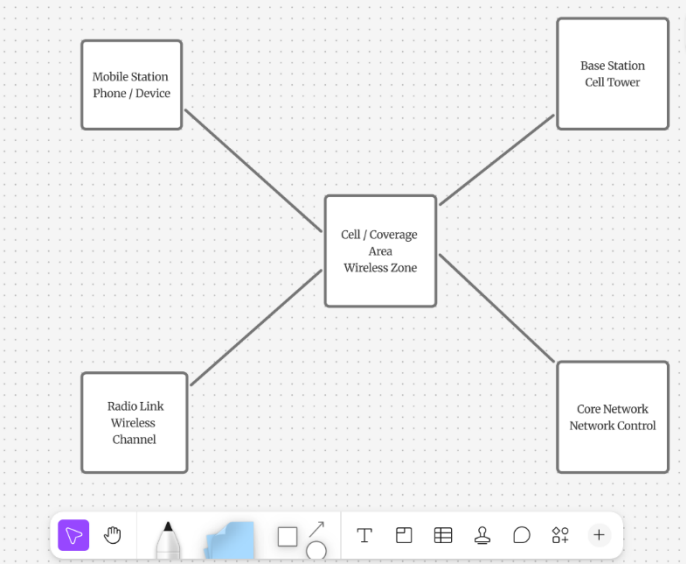
Experiment 1: Basic Cellular System Diagram in Figma

Aim: To create a diagram illustrating the components of a basic cellular system using Figma.

Procedure:

1. Open Figma and create a new design file.
2. Create a frame with a light-gray background.
3. Draw five rectangular boxes and arrange them vertically.
4. Add the names: Mobile Device, BTS, BSC, MSC, and External Networks.
5. Give each box a different color.
6. Connect the boxes using arrows to show communication.
7. Add short descriptions inside each box.

Output:



Result: A basic cellular system diagram was created successfully using Figma.

Experiment 2: Mobile Station UI Design Using Figma

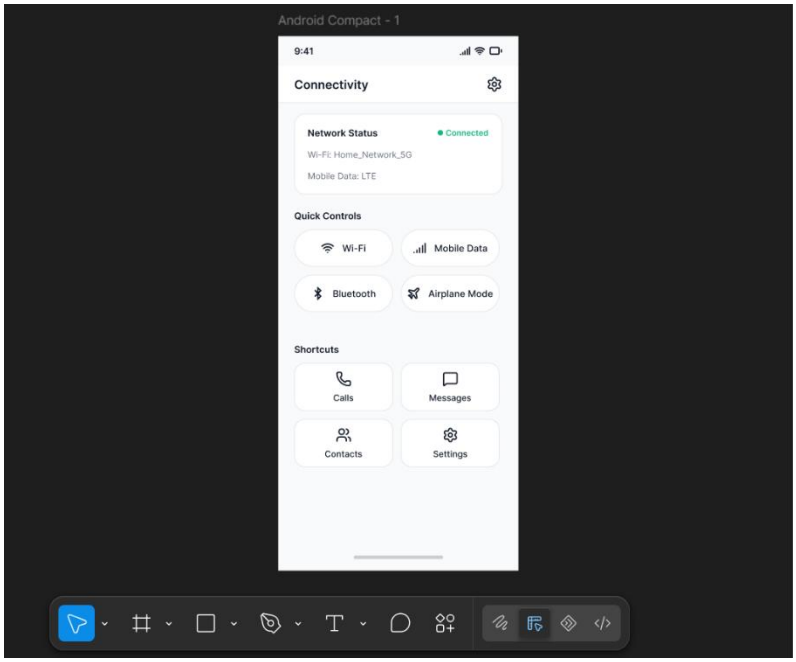
Aim

To design a user interface prototype for a mobile station using Figma, focusing on connectivity and convenience features such as Wi-Fi, mobile data, Bluetooth, calls, messages, and network settings.

Procedure

1. Open Figma and create a new design file.
2. Create a smartphone frame using a suitable mobile device preset.
3. Design the home screen with network status, signal strength, and battery level.
4. Add buttons for Wi-Fi, mobile data, Bluetooth, and airplane mode.
5. Add shortcuts for calls, messages, contacts, and settings.
6. Use suitable colors, icons, and rounded buttons to make the interface user-friendly.
7. Open the Prototype tab and connect the buttons to different screens or interactive states.
8. Test the prototype using Present mode to verify the interactions.

Output:



Result: A basic cellular system diagram was created successfully using Figma.

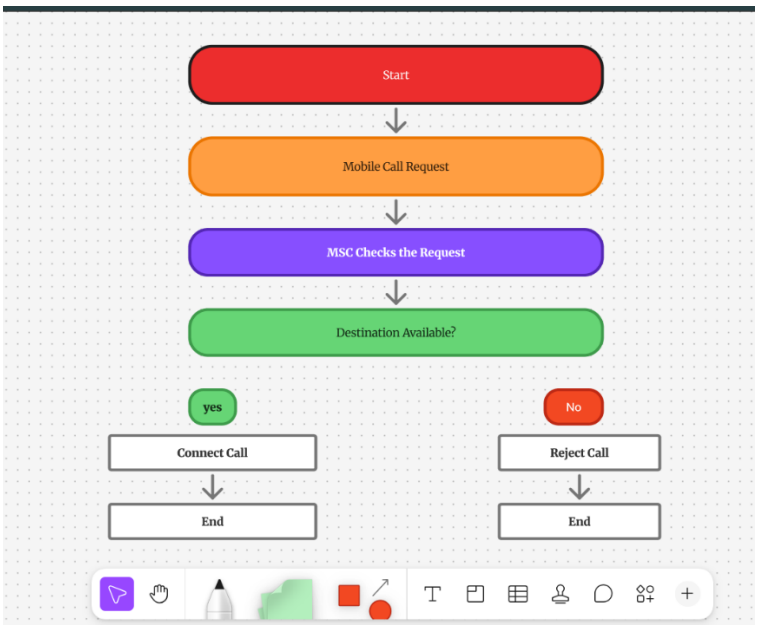
3.Experiment: Network Switching Subsystem Flowchart Using Figma

Aim: To develop a flowchart in Figma illustrating the operation of the Network Switching Subsystem (NSS) in a mobile cellular system.

Procedure:

1. Open Figma and create a new design file.
2. Use oval shapes for Start and End.
3. Use rectangles for processing steps and diamonds for decisions.
4. Connect all shapes using arrows.
5. Add Yes and No labels to the decision branches.
6. Apply suitable colors and align the shapes neatly.
7. Test the flowchart and save the design.

Output:



Result: A flowchart illustrating the operation of the Network Switching Subsystem in a mobile cellular system was created.

4 Experiment Name: Mobile Network Protocol Stack Visualization Using Figma

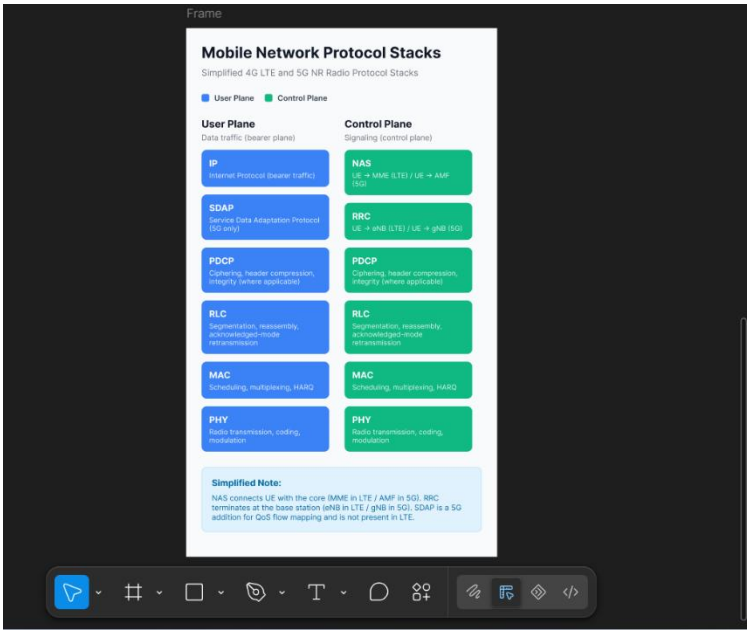
Aim:

To create a visual representation in Figma highlighting the key layers and functions of mobile network protocol stacks.

Procedure

1. Open Figma and create a new FigJam board.
2. Draw five rectangles and arrange them vertically.
3. Label the rectangles as Application Layer, Transport Layer, Network Layer, Data Link Layer, and Physical Layer.
4. Add the main functions of each layer below its name.
5. Use different colors to distinguish the layers.
6. Connect the rectangles using arrows to show the communication flow.
7. Align the shapes neatly and save the design.

Output:



Result:

A visual representation of mobile network protocol stacks was created in Figma, showing the five layers and their main functions in mobile communication.

5 Experiment Name: Mobile Application Wireframe Design Using Figma

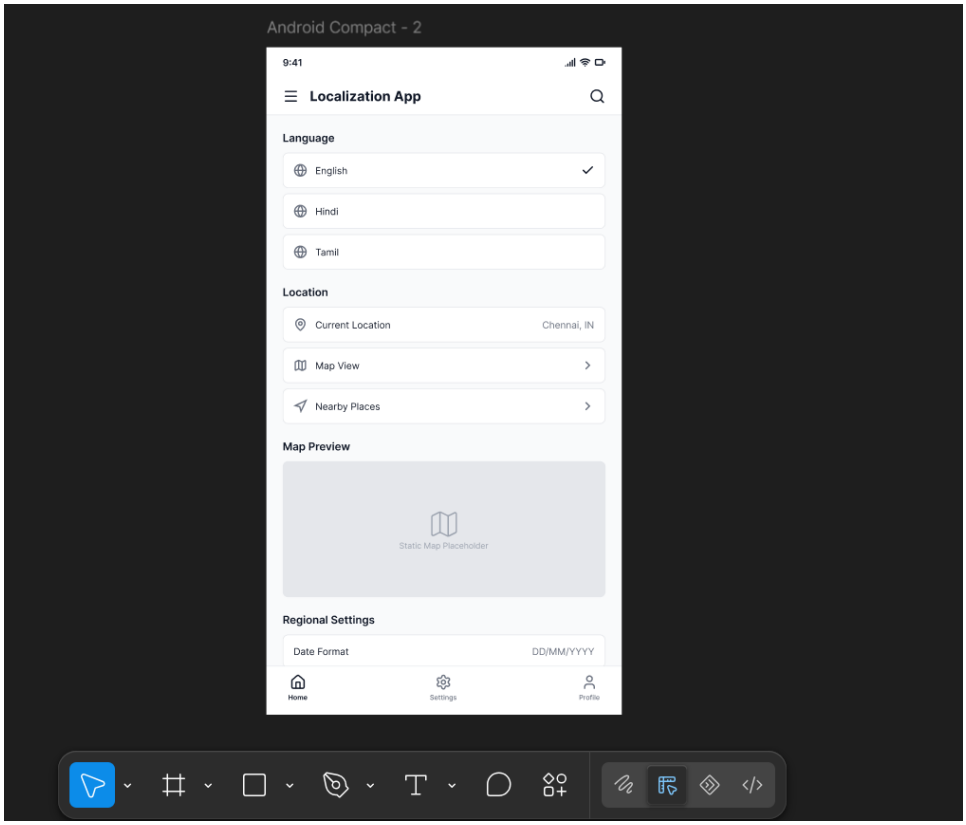
Aim

To design a wireframe for a mobile application using Figma that incorporates localization features such as language selection, regional settings, and location-based services.

Procedure

1. Open Figma and create a new Design file.
2. Select a mobile phone frame.
3. Design a home screen with a title, menu, and navigation buttons.
4. Add a language selection option with choices such as English, Hindi, and Tamil.
5. Add location-based features such as current location, map, and nearby places.
6. Include regional settings such as date format, time format, and currency.
7. Use simple rectangles, text, and icons to create the wireframe.
8. Use the Prototype tab to connect the screens and test navigation.
9. Save the design.

Output:



Result:

A mobile application wireframe was designed in Figma with localization features, including language selection, regional settings, and location-based services.

Link: https://www.figma.com/design/w4E1AZxh5zW76OC0oLS0kk/Untitled?node-id=0-1&p=f&t=IWNqw5Fr4QRvSRDu-0

Link: https://www.figma.com/board/I6rKGZCGbWvS1XafdcFIFo/Untitled?t=IWNqw5Fr4QRvSRDu-0